A self-evolving materials database that unifies 54 public sources into one searchable space of 13.5 million crystals

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Abstract

Crystal structures are not scarce; a materials space that can be searched, compared and traced back to its source is, and assembling one from the public databases consumes effort that should go into screening. They share no key at the level of the object they describe, so one crystal appears in each under a different accession number, and a union merged once ages immediately because every source keeps growing. Here we build that space. Identity is resolved at the level of the crystal across 54 databases, every value keeps the source, record and method that produced it, and admission and recognition are made one operation, so a record matching nothing becomes the corpus's next member and at once enlarges the criterion by which the following record is judged: the database evolves as it ingests rather than being rebuilt. Assembly enlarged the deduplicated pool by 64.7%, and 15.9% of all cross-source matches land on a crystal it had itself added. The result holds 13.5 million distinct crystals and 121.6 million values over 166 properties. Redundancy is large and concentrated, with 1,190,447 crystals described by two or more sources, and those sources often disagree: of 870,683 with a formation energy from more than one, only 60.5% agree within 50 meV atom^{-1}, as a systematic offset rather than scatter. Coverage follows computational cost rather than scientific value, no property needing its own calculation reaching a fiftieth of the crystals, so screening the known record today ranks the materials somebody chose to compute rather than the materials that exist. Predicting the missing entries, for which the corpus supplies 497 bounded learning tasks, would measure the limit of forward screening; the same standardised structures and labels are what generative inverse design requires.

Keywords: materials databases, data integration, crystal structure deduplication, provenance, cross-database benchmarking, machine learning for materials

1 Introduction

Materials discovery has become an exercise in reusing the collective published record. Public repositories now hold millions of computed crystals [1–4], generative models propose new ones directly [5–7], and the universal interatomic potentials used to filter those proposals are trained on whatever the community has already published [8–12]. Crystal structures, in other words, are not scarce.

What is scarce is a materials space that can be searched, compared and traced back to its source in one place. In current practice a screening campaign begins by requesting accounts one repository at a time, calling bulk interfaces, downloading archives in incompatible formats, and then working out which of the resulting records are genuinely distinct crystals rather than duplicates of one another, intermediate relaxation frames or frames of a molecular-dynamics trajectory. Property tables arrive detached from the structures they describe, units and conventions differ between providers, and the licence position of each source has to be established again before anything can be redistributed. The data already exists; a large share of the effort goes into acquiring and tidying it before any candidate is screened.

The reason this preparation cannot be done once and shared is that the repositories agree on no key at the level of the object they describe. The same arrangement of atoms carries a different accession number in every database that holds it, and the one property they genuinely share, the chemical formula, cannot serve as a key, because polymorphs share a formula and differ in almost everything a screening campaign cares about. Without a key at the level of the crystal, the union cannot be deduplicated, values from different providers cannot be brought onto the same object, and three quantities that bear directly on how materials data is used stay unmeasured: how much of the corpus is the same crystal counted repeatedly, whether two groups that computed the same quantity for the same crystal obtained the same answer, and how the properties a discovery campaign actually needs are distributed over the structures that exist.

Assembling such a union once would in any case not be enough. Every repository publishes new releases, and new repositories appear faster than any is retired, so a union built in a single pass is out of date before it is finished and has to be rebuilt from nothing to absorb what arrived while it was being made. A corpus that is to stay current has to do something a merge does not: recognise material it has never seen, and fold it in without discarding the ability to recognise that it has already acquired.

Substantial progress has been made on the parts of this problem that can be solved one layer at a time. The OPTIMADE specification gave more than twenty providers a common query interface,

removing the burden of a separate account and format for each [13, 14]. Structural fingerprints made deduplication tractable at scale, and the LeMaterial initiative used one to publish a deduplicated union of several large repositories [15]. What neither delivers is the property layer, which is where the remaining difficulty sits. Knowing that two records describe the same crystal does not attach what each source says about it, and attaching those statements is useless unless they stay apart afterwards. Two repositories publishing a column labelled band gap are frequently not reporting the same physical quantity: a semilocal, a hybrid and a measured value differ by more than the differences a screening campaign is trying to resolve, so a union built by matching column names would average them and destroy exactly the information that makes the corpus worth consulting.

Here we build that space. Identity is resolved across 54 databases at the level of the crystal structure, and every resulting property value retains the source, the native record and the calculation or measurement that produced it, so that several sources speaking about one crystal remain separately audible. Licence boundaries are enforced in the pipeline rather than by convention, so that the publishable corpus and the locally buildable one are distinct by construction. And admission and recognition are made the same operation: a record that matches nothing is not an error to be logged but the corpus's next member, and the moment it is admitted its structural fingerprint joins the criterion against which every subsequent record is judged. The database therefore evolves as it ingests, and a newly released version or an entirely new repository is folded into a substrate that has grown since the last one rather than matched against a frozen reference.

The space so built answers the three questions its absence had left open, and the answers are not what a user of any single repository would assume. Redundancy is large and strongly concentrated. Independent sources reporting the same quantity for the same crystal disagree by margins exceeding the thresholds on which materials selection routinely turns, and a substantial part of their apparent agreement is shared provenance rather than independent confirmation. Coverage of the property space is governed by what a property costs to compute rather than by what it is worth knowing, which turns the remaining gap into a learning problem the corpus is itself equipped to pose.

That last point decides what the space is ultimately for. Materials discovery proceeds along two routes: forward screening, which searches the known set for a material that already has the property wanted, and inverse design, which generates a structure to order. Forward screening is usually described as bounded by the size of the known set, but that is not what bounds it now. The known set holds 13.5 million distinct crystals, and the best-covered property that requires a calculation of its own reaches fewer than one in fifty of them, so nearly every known material

has never been evaluated for nearly every property and a screen returns not the best material but the best characterised one. Completing the property layer over the whole corpus therefore does something that merely enlarging the corpus would not: it establishes what forward screening can actually deliver from the record as it stands, and how much of that record has been passed over for want of a calculation rather than for want of merit. Whatever the completed screen fails to find is what inverse design has to supply, and the same corpus is what a generative model needs in order to try, since it provides deduplicated structures against which novelty can be judged across the whole known record rather than one repository, and property labels that name the quantity they denote. The instrument and the corpus it produces are described in Methods and in the Supplementary Information; what follows is what the corpus turns out to contain.

2 Results

2.1 One searchable space assembled from 54 sources

We resolved identity across 54 public databases at the level of the crystal structure, matching each incoming record first on a cross-database identifier the source is entitled to assert and, failing that, on a structural fingerprint, never on the chemical formula (Fig. 1a; Methods). Records that match nothing are admitted rather than discarded, and every value retains the database, the native record and the method behind it.

The corpus so assembled holds 13,543,277 distinct crystal structures, of which 357,127 are experimental and 13,186,150 are computational candidates. On that pool sit 121,620,776 property values spanning 166 properties, 1,198,897 stored spectra and 17,327,895 references recording which database contains which crystal.

The contributors are extremely unequal (Fig. 1b). Their contributions span 5.7 orders of magnitude; the largest supplies half of all values and the five largest supply four fifths, while 30 of the 54 supply fewer than 10^5 each. The asymmetry is not a curiosity of bookkeeping, because the two ends of the distribution do different work. The large general-purpose databases determine how much of the pool carries any property at all; the small specialised libraries determine how many distinct properties exist to carry, and the single most property-diverse contributor supplies 61 of the 166 properties while the largest supplies 10.

Assembling the corpus enlarged it (Fig. 1c). The deduplicated starting pool held 8,225,138 structures; admitting the records that matched nothing added 5,318,139 more, a growth of 64.7%.

Of the 17,327,895 references, 6,784,564 were resolved by record identifier, 5,126,030 by structural fingerprint alone, and the remainder record a database's own entry (Fig. 1d). The two matched routes are not interchangeable. Identifier matching is exact but available only where a database publishes a cross-reference it owns, which is why it dominates among the general-purpose repositories and is nearly absent among the framework libraries; fingerprint matching is available everywhere but has a domain of validity.

That domain has an edge, and it is worth stating before any conclusion is drawn from the corpus. A bonding-graph fingerprint identifies a crystal by its connectivity and its primitive composition, which is the right invariant for crystals that differ in either. Enumerated framework libraries do not: they are generated by decorating a small number of parent topologies, so sibling entries are identical in everything the fingerprint reads. Left untreated, the fingerprint path collapses more than 10^5 distinct hypothetical zeolite frameworks onto a few hundred pool materials and discards a comparable number of hypothetical metal–organic frameworks as duplicates of one another. We detect this at the level of the source rather than the record, and admit such libraries under a composite key that adds the cell volume per site, an invariant the graph hash ignores (Supplementary Note S2). An audit that checks each source independently for load integrity, staging yield and field coverage reports no discrepancies over the completed database (Supplementary Note S4).

A structure admitted from one database becomes recognisable to every database that follows, because its fingerprint and identifiers join the matching substrate as it is admitted. The corpus therefore does not merely accumulate: the criterion by which it recognises the next record is itself a product of the records already ingested, and it is by this mechanism that the database keeps itself current as sources publish new versions and new sources appear. The effect is measurable. Of the 11,910,594 references resolved by matching, 1,898,275, or 15.9%, land on a crystal that was absent from the starting pool and was admitted during assembly (Fig. 2c). Those links do not exist in a construction that matches every database against a fixed reference.

The effect is ordered by ingestion time and by chemical distance from the starting pool. AFLOW, ingested about ninety minutes after Alexandria contributed 1,514,488 structures, resolved 47.7% of its 1,401,895 matches onto crystals admitted during assembly. The JARVIS tight-binding set, whose fitting geometries are deliberately non-equilibrium and largely absent from the starting pool, reached 67.6%. The framework libraries show the effect in its purest form: CoRE MOF, ingested last, resolved 99.5% of its matches onto frameworks contributed by four earlier libraries in the same run. Databases whose chemistry the starting pool already covered show the converse, at or near

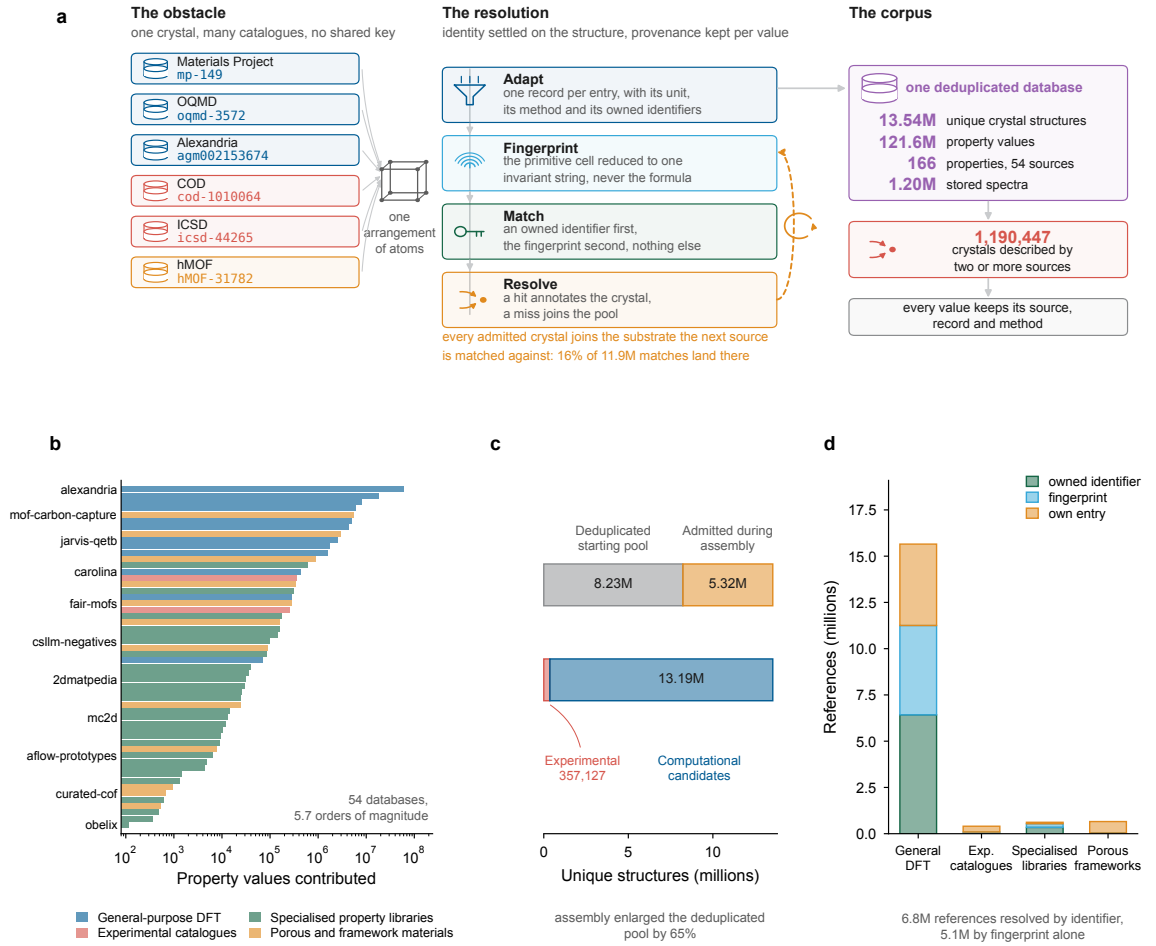


Fig. 1 Making 54 materials databases comparable at the level of the crystal. a, The obstacle and the resolution. Each database labels the same arrangement of atoms with its own accession number, and no key relates them. Every incoming record is reduced to a structural fingerprint and matched first on a cross-database identifier the source is entitled to assert, second on that fingerprint, never on the chemical formula. A record matching nothing is admitted and its fingerprint re-enters the matching substrate, so each database is matched against a larger pool than the one before it. The right column reports what the assembled corpus holds. **b**, Property values contributed by each of the 54 databases on a logarithmic axis, coloured by family, with labels at regular intervals through the ranking. The contributions span 5.7 orders of magnitude and the largest database supplies half of all values. **c**, Composition of the structure pool. The upper bar divides it into the deduplicated starting pool and the structures admitted during assembly; the lower bar divides the same total into experimental records and computational candidates. **d**, How the references were resolved, grouped by database family. Identifier matching dominates where a database publishes cross-references it owns; fingerprint matching is available everywhere; own entry records that the database contains the structure itself.

zero. The practical consequence is that the marginal value of adding a database is not confined to the values it brings: CoRE MOF contributed 155,488 experimental porosity measurements, and the reason 99.5% of them landed on an existing crystal rather than creating an isolated set of new ones is that earlier libraries had already put those frameworks in the pool. Assembled against a frozen reference, the same measurements would have arrived as an unconnected island.

2.2 Redundancy between databases, and whether the duplicates agree

How much of the public corpus is the same crystal counted more than once is a question that only a structurally resolved union can answer, and the answer is that a substantial fraction is. Of the 12,827,418 structures carrying at least one database reference, 1,190,447 carry two or more (Fig. 2a). The distribution falls steeply and reaches far: 870,439 crystals are described by exactly two databases, 158,623 by three, 2,058 by thirteen and one by 24. Redundancy is not spread evenly over chemistry; it is concentrated where the field has concentrated its effort. The largest single overlap holds 451,178 shared crystals, and the ten largest overlaps are all between general-purpose density functional theory repositories (Fig. 2b). These are the numbers a federated query cannot report and a formula-level join would misstate in both directions, and they bound what any cross-database evaluation can claim: for the largest overlapping pairs, a model trained on one repository and tested on another is tested on hundreds of thousands of crystals it has already seen, unless the split is taken over structures.

Redundancy is only interesting if the redundant copies can be compared, and they can be compared only because every value kept the method that produced it. Where two databases report the same quantity for the same crystal under comparable methods, we took the median of each database's own values and the spread across databases (Fig. 2d–f; Supplementary Note S7). The comparison is possible on a large sample: 870,683 crystals carry a formation energy from two or more databases, 452,584 a band gap, 376,813 a hull distance, 145,353 a magnetisation and 15,275 a bulk modulus. Concordance falls as the quantity becomes harder to compute. Within 50 meV, 75.9% of shared band gaps agree; within 50 meV atom^{-1}, 64.1% of hull distances and 60.5% of formation energies do; within 5 GPa, 44.2% of bulk moduli and 49.4% of shear moduli do.

The tail matters more than the median. The tenth of shared formation energies that agree least differ by more than 248 meV atom^{-1} (Fig. 2e), which is several times the hull-distance margins conventionally used to call a compound stable [16, 17]. Whether a candidate passes such a filter can

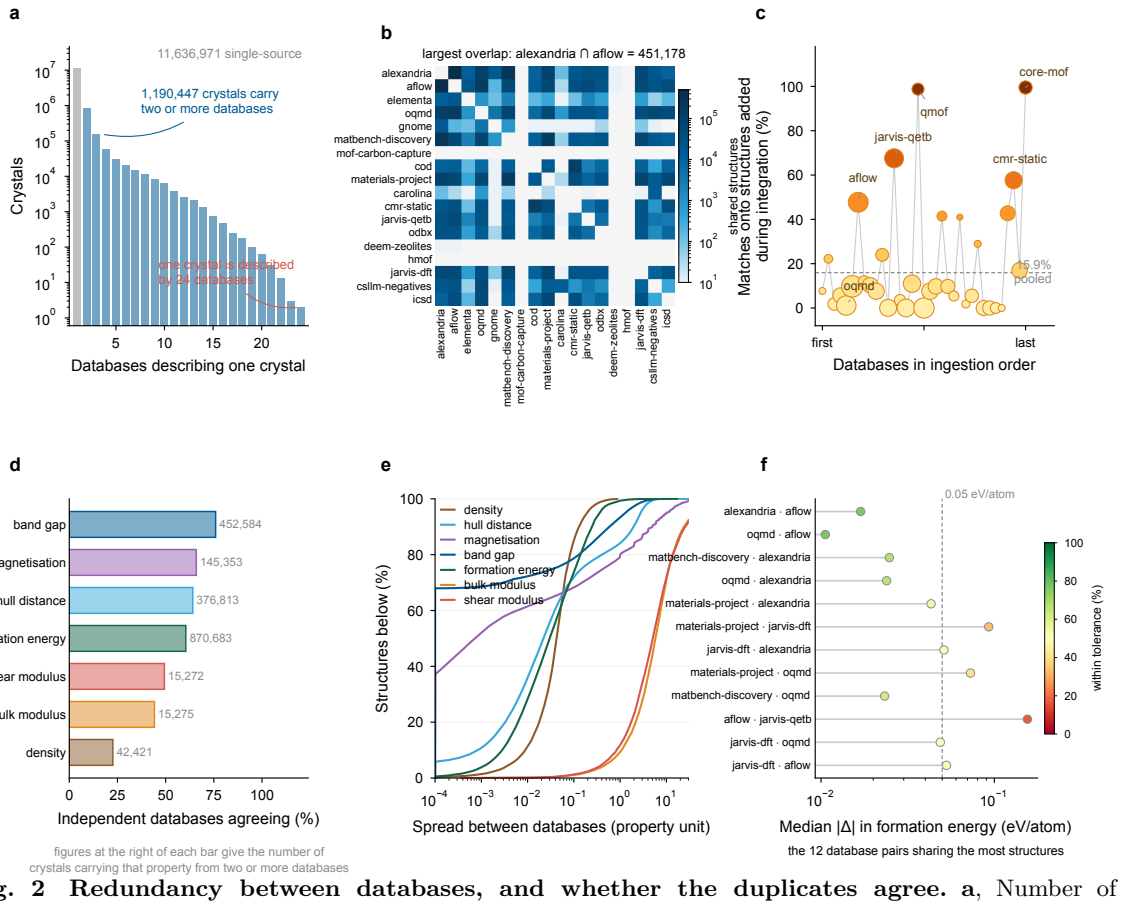


Fig. 2 Redundancy between databases, and whether the duplicates agree. a, Number of databases describing one crystal. Grey marks the 11,636,971 crystals held by a single database; blue marks the 1,190,447 held by two or more. The distribution reaches 24. **b**, Shared crystals between the 18 databases annotating the most structures, on a logarithmic colour scale. Empty rows are not missing data: the hypothetical-framework libraries share almost nothing with the general-purpose repositories. **c**, The growing recognition substrate. For each database with at least 1,000 matched references, in ingestion order, the fraction of its matches landing on a crystal absent from the starting pool and admitted during assembly. Marker area scales with the number of matches; the dashed line is the value pooled over all databases. **d**, For each comparable property, the fraction of multi-source crystals whose independent values fall within the tolerance stated in Supplementary Note S7; the figure at the right of each bar is the number of crystals carrying that property from two or more databases. **e**, Cumulative distribution of the spread between databases, in the property's own unit for energies, magnetisation and moduli and as a relative difference for density. **f**, Median absolute difference in formation energy for the twelve database pairs sharing the most crystals, coloured by the fraction agreeing within 50 meV atom^{-1}. Pairs at the left agree because they carry the same underlying calculation; pairs at the right differ by a systematic offset.

therefore depend on which database supplied its energy, and in a corpus without structural identity that dependence is not merely unquantified but unobservable.

Resolving the comparison by pair separates two regimes that a single agreement figure would merge (Fig. 2f). Some pairs agree almost exactly. Two of the largest repositories share 138,961 crystals whose band gaps have a median absolute difference of exactly zero and agree within tolerance for 93.3% of them. That is not independent confirmation; it is the signature of shared provenance, two datasets carrying one underlying calculation, and reading it as corroboration would invert its meaning. Other pairs disagree systematically. For formation energy the median signed difference between the Materials Project and JARVIS-DFT is $+70.7$ meV atom^{-1} against a median absolute difference of 92.7 meV atom^{-1}, so nearly all of the disagreement carries one sign; the corresponding figures are $+171$ against 177 meV atom^{-1} for the Materials Project against AFLOW. Pearson correlations across these pairs remain between 0.85 and 0.99, so what separates the databases is an offset and a scale rather than noise. This is precisely the structure that merging columns by name would absorb without trace, and precisely the structure a model trained on the merged column would learn as if it were physics. Method strings make the largest offsets interpretable rather than merely visible: the widest gap among the major pairs, $+150$ against 155 meV atom^{-1}, is between a density functional and a tight-binding dataset, and is legible as such only because every value in the latter carries a method string that says so.

Holding several databases against one another also exposes individual values that no single-source check would question. Among the shared elastic moduli, 380 bulk moduli published by one repository and 316 published by another lie outside any physically plausible range, being negative or above 1000 GPa, with the largest magnitude reaching 4.7×10^{11} GPa. These are unconverged elastic-tensor fits published as they stand. In isolation each is a number in a column; set against a partner value on the same crystal, each is immediately identifiable.

2.3 Coverage follows what a property costs, not what it is worth

The assembled corpus makes its own principal deficiency measurable. Of the 166 properties in the registry, 151 carry at least one value; treating the pool and those 151 properties as a matrix gives 2,045,034,827 cells, of which 95,128,921 are occupied, so the matrix is 4.65% full (Fig. 3a). Occupancy is extremely uneven: nine properties cover more than 40% of the pool, while 95 of the 151 cover less than 1% and 58 cover less than 0.1% (Supplementary Table S2). Half of all crystals carry at most six of the 151 properties (Fig. 3c).

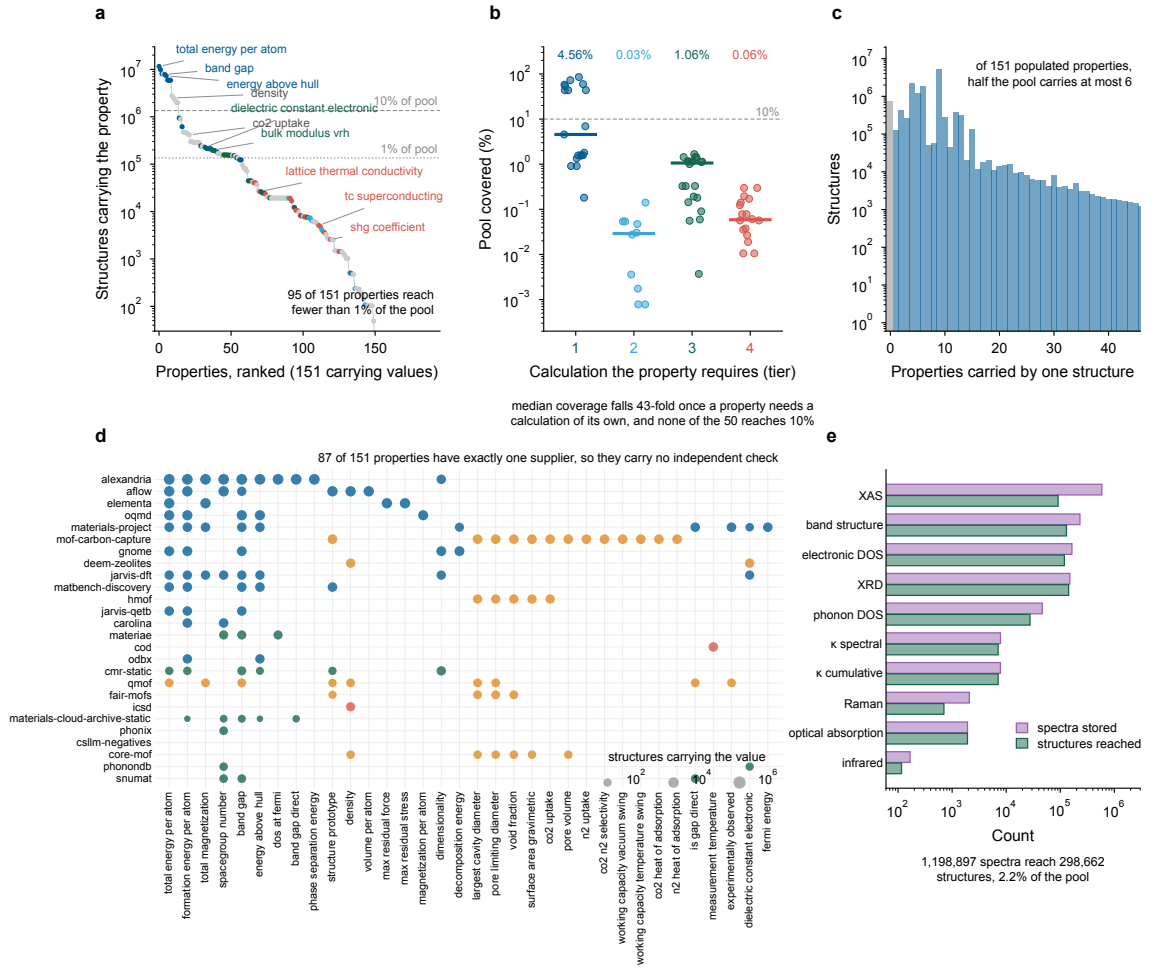


Fig. 3 What the corpus contains, and what it does not. a, The 151 properties carrying at least one value, ranked by the number of crystals that carry them. Colour marks the calculation tier assigned in **b**; grey marks properties outside that classification, whose domain is a single material family. **b**, Coverage of the 69 general-purpose properties grouped by the calculation each requires: tier 1, readable from a single self-consistent run; tier 2, a few extra self-consistent runs such as a slab or a strained cell; tier 3, linear response or finite differences over many displacements; tier 4, anharmonic, transport or beyond-semilocal treatment. Each point is one property, horizontal bars are group medians and the percentages above give those medians. No property in tiers 2 to 4 reaches 10% of the pool. **c**, How many of the 151 populated properties a single crystal carries. The median is six. **d**, Which database supplies which property, for the 34 most widely covered properties and the 26 databases annotating the most crystals. Marker area scales with the number of crystals carrying that value, and colour marks the database family. **e**, The spectrum store by curve type: spectra held, and the number of distinct crystals they reach. X-ray absorption spectra are the most numerous and among the least widely spread, because one crystal carries one spectrum per absorbing element and edge.

The gradient does not follow scientific importance. It follows price (Fig. 3b). Grouping 69 general-purpose properties, excluding those whose domain is one material family, by the calculation each requires (Supplementary Note S8), 19 are read off a single self-consistent calculation a database performs anyway: total and formation energies, hull distance, band gaps and edges, magnetisation, the density of states at the Fermi level. Their median coverage is 4.56% of the pool and eight exceed 10%. The remaining 50 require a calculation of their own, from an extra slab or strained cell through linear response to anharmonic phonons, transport integrals and hybrid functionals. Their median coverage is 0.107%, lower by a factor of 43, and *not one of the fifty reaches 10%*. The best covered among them is the electronic dielectric constant at 1.67%, followed by the Voigt–Reuss–Hill bulk modulus at 1.45%.

Within that second group the ordering is set by whether a dedicated campaign ever existed rather than by cost alone. Linear-response properties, the more expensive of the remaining tiers, have a higher median coverage (1.06%) than the properties reachable from a few extra self-consistent runs (0.029%), because the elastic and dielectric tensors were the explicit target of high-throughput efforts [18, 19] while quantities such as the work function were not. Coverage in this corpus therefore records where the community chose to spend computer time, and that map is legible only once the databases sit on one structure pool.

The same stratification governs non-scalar data and provenance. The curve store holds 1,198,897 spectra across ten types but reaches only 298,662 crystals, 2.2% of the pool (Fig. 3e). And only 64 of the 151 populated properties are reported by two or more databases; the other 87 have a single supplier and therefore no independent check of any kind. Of the 95,128,921 occupied cells, 3.9% carry values from two or more sources, which is the exact extent of the corpus on which the comparison of Section 2.2 is possible.

2.4 Learning the properties that were never computed

The coverage gradient states a problem the corpus is now equipped to attack. Nearly ten million crystals carry a formation energy and 7.8 million a band gap; 196,615 carry a bulk modulus and 26,507 a lattice thermal conductivity. The missing entries are not missing because they are uninteresting. They are missing because obtaining them costs between one and four orders of magnitude more than the calculation that produced the entries that are present, and filling the matrix by computation would require an effort no single group can supply. Learning it instead requires three

things a single database does not provide, and that the assembled corpus does: deduplicated structures, so that a train–test split falls on distinct crystals rather than on records; labels that retain the quantity they denote; and, on the subset with two independent sources, a measurement of the noise floor a prediction should be compared against.

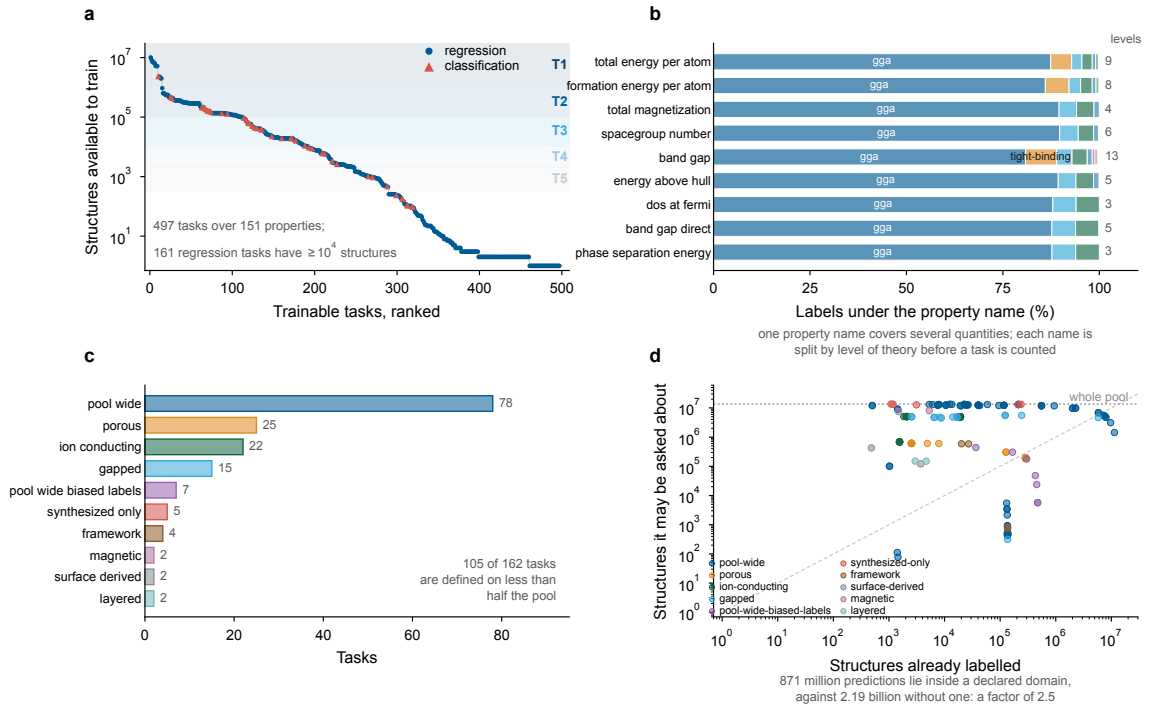


Fig. 4 What the corpus offers a learned completion of the property layer. a, Every candidate task, defined as one property at one level of theory, ranked by the number of crystals available to train it. Shaded bands mark the data tiers T1 to T5 at 10^6, 10^5, 10^4, 2×10^3 and 3×10^2 crystals. Circles are regression targets, triangles classification targets. **b**, Composition of the nine largest properties by level of theory, as a fraction of the labels filed under each property name; the figure at the right gives the number of distinct levels. Band gap carries 13, of which the semilocal group holds 81%. **c**, The 162 constructed training sets by declared applicability domain, the set of structures for which the property is defined at all. **d**, For each training set, the crystals already labelled against the crystals the task may legitimately be asked about once its domain is enforced. The dotted line marks the whole pool and the dashed diagonal marks parity. Points far above the diagonal are the tasks with the most to gain; points on the right at low height are tasks whose domain is already almost fully labelled. Panels reporting the trained models are added when their held-out numbers exist.

A census of the database establishes how much is actually trainable (Fig. 4a). The unit that matters is not the property but the property at one level of theory, because a name such as band gap covers several different physical quantities that a model cannot distinguish from structure alone. Split that way, the 151 populated properties yield 497 candidate tasks. Of these, 161 regression tasks have at least 10^4 crystals to train on, 97 have at least 10^5, and 13 have at least 10^6. Constructing the training sets under a default policy that keeps the largest level of theory per property produced 162 sets covering 116 properties and 93,462,873 labels, each with a chemistry-aware train, validation and test split.

Three properties of the corpus determine how those labels must be handled, and each of them is a direct consequence of the integration reported above. The first is that a property name is not a label definition (Fig. 4b). Band gap carries 13 distinct levels of theory, of which the semilocal group holds 81% and a tight-binding fit a further 8%; the systematic offsets between them are of order 1 eV, which a model cannot infer from the structure and would instead learn as a proxy for which database supplied the row. The second is that some quantities are condition-dependent rather than multiply determined: carbon dioxide uptake is reported at four distinct grand-canonical conditions and hydrogen uptake at five temperature and pressure combinations, so a material legitimately carries several different correct values and taking their median produces a number that is not any physical quantity. The third is the cross-database disagreement of Section 2.2, which here becomes an operational quantity rather than an observation: across the 162 training sets, 5,831,192 crystals carry labels from two or more databases and 519,599 of them disagree by more than a quarter of the target's standard deviation. Those spreads are retained alongside the resolved label so that they can weight the loss, rather than being averaged away.

Splitting the data is a fourth constraint of the same kind. The same compound recurs in the pool under different accession numbers, so a random split places identical chemistry in training and test, and the resulting scores are optimistic in exactly the regime that matters, since the crystals a completion is wanted for are the ones whose chemistry the training set never saw. Holding out whole chemical systems is the right default, but it degenerates on the enumerated libraries, where an entire task can occupy a single chemical system: a chemistry-level holdout leaves the validation and test folds of such a task empty, which happened for 23 of the 162 sets before the split rule was made to fall back and to record which level it used.

Finally, the completion has a domain (Fig. 4c,d). Carbon dioxide uptake is a grand-canonical adsorption quantity: it is not zero for a dense intermetallic, it is undefined, because the simulation cell has nowhere to insert a molecule. Of the 162 training sets, 63 draw more than 90% of their labels from a single material family, and the largest declared domains are pool-wide (78 tasks), porous (25), ion-conducting (22) and gapped (15). Enforcing those domains changes the size of the completion problem substantially: 105 of the 162 tasks are defined on less than half the pool, and the number of predictions that lie inside a declared domain is 871,097,089 rather than the 2.19 billion that asking every task about every crystal would produce, a factor of 2.5. The difference is not a loss of coverage. It is the removal of numbers that would have looked like data.

One dependency in that structure is worth stating because it closes the loop with Section 2.1. Seventeen of the tasks are gated on a property the database does not yet hold: whether a crystal has a gap, or whether it is magnetically ordered, is unknown for 75,674,942 structure-task pairs, so their admissibility can only be decided once the corresponding tier-1 models have run. The completion therefore proceeds in stages, and each stage enlarges the domain over which the next one is defined, in the same way that each admitted structure enlarged the substrate against which the next database was matched.

What such a completion would settle is worth stating before it is attempted, because it is the reason the corpus was assembled rather than merely enlarged. Screening the known set for a material that already has the property wanted is bounded, at present, not by how many crystals are known but by how few of them have been characterised. Fewer than one crystal in fifty carries even the best-covered property that needs a calculation of its own, and the median such property reaches one in a thousand (Section 2.3), so a screen conducted today over the public record ranks the materials somebody chose to compute rather than the materials that exist. Predicting the missing entries over the whole corpus, within the domains of Fig. 4c,d, converts that ranking into something closer to a complete one, and thereby measures a quantity the field has never had: how much of the known record has been passed over for want of a calculation rather than for want of merit. A large yield would mean that forward screening has been operating far inside its own reach; a small one would mean the reach is nearly exhausted and that further progress has to come from designing structures rather than finding them. The same corpus serves that second route as well, since a generative model needs exactly what it holds: structures deduplicated across the whole known record, so that novelty is judged against everything published rather than against one repository, and property labels that name the quantity they denote, so that a model can be conditioned on a defined target. Which of the two answers the completed layer gives is an empirical question, and it is the one the next stage is built to decide.

[In progress: the trained models]

The models themselves are still training. This block becomes the closing passage of Section 2.4 and adds panels to Fig. 4: the multi-task backbone over the thirteen tier-1 tasks and the fine-tuned heads over the smaller ones, held-out accuracy per task set against the multi-source noise floor of Section 2.2, the coverage the completed layer reaches for the properties of Fig. 3b, and the boundary within which a predicted entry may be used. It is written when the held-out numbers exist, so that no claim precedes its evidence.

3 Discussion

What changes when the public record is held as one structurally resolved space is first of all practical. A screening campaign that used to begin with account requests, bulk downloads and format conversion begins instead with a query, and the answer carries with it the source of every number in it. A crystal is one row rather than six, so counting is possible; a property value names the calculation behind it, so comparison is possible; and the licence position of every value is a column rather than a matter to be re-established. This is infrastructure rather than discovery, and its value is measured by what it removes from everyone else's work.

It also makes a class of question answerable that was not. A repository can report on its own contents; only a resolved union can report on the field. How many distinct crystals are actually known, as opposed to how many records exist. Which groups have already computed a given compound, and whether they agreed. Whether a training set and a test set drawn from two providers share the same crystals. None of these is a property of any member, and none can be recovered afterwards from a corpus assembled by concatenation.

Two of the answers were not what we expected, and both bear on how the corpus should be used. Independent sources reporting the same formation energy for the same crystal agree within 50 meV atom^{-1} only 60.5% of the time, and the least concordant tenth differ by margins several times the thresholds on which stability screening turns. Because the discrepancy is a systematic offset rather than scatter, it does not average away with sample size, and a corpus that merged the contributing columns would hand a model the offset to learn as though it were physics. The remedy is not to choose a preferred provider but to keep the provenance, which costs one column per value and turns the disagreement into an observable. The complementary caution is equally practical: a pair of sources agreeing to the last decimal is usually carrying one calculation twice, so agreement between repositories should be read as evidence of shared provenance until shown otherwise. The second answer concerns what the corpus does not contain. Coverage of the property space follows the price of the calculation, not the value of the answer, and within the expensive tier it tracks whether a dedicated high-throughput campaign happened to exist. The resulting landscape is a record of where the community spent computer time, which is a different thing from a record of what is worth knowing.

That landscape is also a well-posed learning problem, and the space supplies what such a problem needs: 497 candidate tasks once each property name is split by level of theory, 162 of them realised as training sets over 93 million labels, deduplicated structures to split on, labels that retain the

quantity they denote and, on the entries with two independent sources, a measurement of the floor below which a prediction cannot be distinguished from the disagreement of its own training data. Section 2.4 also shows how easily that problem is posed wrongly. Roughly two thirds of the tasks are defined on less than half the corpus, and asking each of them about every crystal would generate 2.19 billion numbers of which more than half would be undefined rather than merely uncertain. Enforcing the domains is not a loss of coverage; it is the difference between a completed property layer and a plausible-looking one.

The reason to complete it goes beyond adding numbers to a table. Materials discovery runs on two routes, and the state of the public record has kept them apart. Forward screening searches the known set for a material that already has the property wanted; inverse design generates a structure to order. Forward screening is habitually treated as bounded by how many materials are known, but the corpus shows that it is bounded instead by how few of them have been characterised: no property requiring a calculation of its own reaches a fiftieth of the 13.5 million crystals, so a screen conducted over the public record ranks the materials somebody chose to compute rather than the materials that exist, and its apparent frontier is an artefact of where computer time was spent. Predicting the missing entries across the whole corpus, inside the domains where each property is defined, replaces that artefact with something closer to the real frontier and answers a question the field has not been able to pose: how much of the known record has been passed over for want of a calculation rather than for want of merit. The two possible answers are both consequential. If the yield is large, forward screening has been operating far inside its own reach and a substantial number of already-synthesisable materials are waiting to be recognised. If it is small, the reach of the known set is close to exhausted, and further progress has to come from designing structures rather than finding them. The same corpus serves that second route, because a generative model needs what it holds: structures deduplicated across the whole published record, so that a claim of novelty is tested against everything known rather than against one repository, and property labels that name the quantity they denote, so that generation can be conditioned on a defined target rather than on a column heading. Standardising structure and property data at this scale is what makes the two routes comparable, and eventually complementary, rather than separate literatures.

One property of the construction is worth separating from the results it produced. Admission and matching are the same mechanism seen from two sides: a record that matches nothing is not an error to be logged but the corpus's next member, and once admitted it becomes part of the criterion by which the next record is judged. Over the full assembly, 15.9% of matches resolved onto crystals

that assembly itself had added, approaching unity for chemistries the starting pool did not cover. The same relation reappears one level up in the completion stage, where seventeen tasks can only be gated once earlier models have supplied the property their domain is defined by. A space built this way has no final state: each round of ingestion enlarges what the next round can recognise, and each round of prediction enlarges the domain over which the next round is defined. That is what makes it maintainable as its sources keep moving, which a merged snapshot is not.

Several limitations bound these conclusions. Structural fingerprinting has a domain of validity, and enumerated framework libraries fall outside it; the composite key rescues the two families we encountered but is a remedy for a known failure rather than a general solution. Agreement can be measured only where two providers report the same quantity under comparable methods, which the method strings make checkable but which holds for a minority of the occupied matrix, so the disagreement reported here is measured on the part of the corpus most likely to be well behaved. The applicability gate is structural rather than geometric: a porous domain is enforced through family labels rather than a computed pore-limiting diameter, which is the stricter test and remains to be run over the whole corpus. Licence fragmentation persists downstream, since four sources contributing 26.0% of all values cannot be redistributed, so the public artifact and the locally built one are not the same object and any published statistic has to say which it describes. And no value here was recomputed. Every number is as its source reported it, and the union inherits whatever errors those sources contain; what it adds is the ability to see when two of them disagree.

4 Methods

4.1 Sources and their ingestion

Each of the 54 databases is reached through one adapter emitting a stream of records, each carrying the source's native identifier, its property values with units and methods, its structure where the licence permits redistribution, and any cross-database identifiers the source is authoritative for. Sources that may not redistribute structures are marked properties-only in code, so no structure of theirs can enter the pool. The complete inventory, with each database's licence, contribution and reference routes, is Supplementary Table S1; the adapter protocol and staging pipeline are Supplementary Note S1.

Property names are mapped onto a canonical registry of 166 entries, each with a fixed unit and value kind. A source field enters the registry under an existing name only when the quantity is the

same physical quantity, and otherwise gains its own name. Method strings are required to distinguish values that a source reports for the same material under different settings, and the loader treats (native identifier, property, method) as the uniqueness key. The registry and its boundary cases are Supplementary Note S3.

4.2 Identity resolution

Structures are fingerprinted with the BAWL hasher [15] under the protocol used for the original deduplicated merge, so that fingerprints computed here are comparable with those already in the pool: parse, reject cells above 500 sites and disordered occupancies, reduce to the primitive cell with pymatgen [20], reject primitive cells above 300 sites, then hash.

Matching proceeds in two ordered paths against a map holding fingerprint-to-material and (source, identifier)-to-material entries. A record is looked up first by every authoritative cross-database identifier it carries, and then by fingerprint. Chemical formula is never a match criterion. A record matching nothing is admitted as a new pool material, and its fingerprint and identifiers enter the map, which is the mechanism measured in Section 2.1. Enumerated framework libraries, whose siblings are indistinguishable to a bonding-graph fingerprint, are detected at the level of the source and admitted under a composite key adding the cell volume per site (Supplementary Note S2).

Every match records the material, source, native identifier and route taken. Values are never overwritten or averaged: several databases reporting one property for one material coexist as separate rows.

4.3 Audit

Completeness is verified by three checks applied to each source independently: the row count the loader reported must equal the row count the database holds; every staged row must be either loaded or accounted for as an identifier no pool material corresponds to; and every property name present in the staged data must be present in the database for that source. The third catches a field the adapter maps but the loader never lands, which leaves row totals plausible while an entire quantity goes missing. The completed database passes all three for all 54 sources (Supplementary Note S4).

Every aggregate statistic reported here is computed with a single grouped scan over the property table. The natural alternative, a correlated subquery per source, rescans 121 million rows 54 times and evaluates each source against a different snapshot.

4.4 Cross-database comparison

Agreement was computed for eight properties whose values are comparable between databases once the method is fixed; extensive and convention-dependent quantities were excluded, because two databases disagreeing on a total energy reflects a different reference state rather than a different answer. Within each material, each source was reduced to the median of its own values, so a source reporting several methods is counted once, and the spread was taken as the range across sources. Tolerances, the excluded quantities and the full pair table are Supplementary Note S7 and Table S4.

The 69 general-purpose properties of Fig. 3b were assigned by hand to four tiers by the calculation each requires, excluding properties whose domain is one material family. The assignment is listed in full in Supplementary Table S3 so that it can be disputed item by item.

4.5 Census and training-set construction

A task is defined as one property at one level of theory, since method strings collapsed onto a single name denote different physical quantities. Under that definition the populated properties yield 497 candidate tasks, and a default policy keeping the largest level of theory per property realises 162 of them as training sets over 93,462,873 labels. Condition-dependent properties are split by condition rather than reduced to a median; cross-database spread is retained beside the resolved label; and splits are taken over chemical systems, falling back through formula to random where a task occupies too few systems to yield non-empty folds. Each task carries a declared applicability domain and an empirical one measured from its training set, and predictions are admitted only inside both. The full protocol, the four constraints it addresses and the domain assignments are Supplementary Note S9.

4.6 Software and reproducibility

The pipeline is implemented in Python; structure handling uses pymatgen [20] and symmetry analysis uses spglib [21]. The database is one SQLite file with two side-car binary blobs holding structures and spectra. Every figure is produced by a script reading comma-separated data files, and each of those files is produced by a named extraction script reading the published database read-only. No number quoted in this work was transcribed by hand.

Acknowledgements. We thank the maintainers of the 54 source databases integrated here. This work redistributes none of their data beyond what their licences permit, and depends entirely on their decision to publish in the first place.

Declarations

Competing interests. The authors declare no competing interests.

Data availability. The unified database is published as a versioned dataset on the Hugging Face Hub together with a development slice sized for interactive work. Four of the 54 sources restrict redistribution: AFLOW's terms are academic and non-commercial, ELEMENTA and GNoME are CC BY-NC 4.0, and Inorganic Crystal Structure Database records may be held only by a subscriber to the FIZ Karlsruhe licence. Those sources are searchable in a database built locally from the published pipeline and are filtered out before any public release. The database contains no ICSD structure. The comma-separated data files behind every figure are provided as source data.

Code availability. The acquisition adapters, the fingerprinting and matching pipeline, the loader, the completeness audit, the extraction scripts and the figure scripts are available in the repository accompanying this work.

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